

Evaluating and Equivalent Expressions

Lesson Plan

1. Lesson Information

Course	Grade 7 Accelerated (1205050)	Unit	3
Lesson	Lessons 6–7 — Evaluating and Equivalent Expressions	Date	
Length	53 minutes (one period)	Benchmark(s)	MA.8.NSO.1.3

2. Standards

MA.8.NSO.1.3 Extend previous understanding of the laws of exponents to include integer exponents. Apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to integer exponents and rational-number bases, with procedural fluency.

Must. Integer exponents and rational bases, with procedural fluency: evaluate, and decide equivalence by rewriting to a common base.

Must not. Bases that get rewritten are powers of 2, 3 or 5. No prime factorization of arbitrary numbers, and no fractional exponents.

3. Mathematical Thinking and Reasoning Standards

the evidence for each, from this period

MTR	What it asks	The evidence in this lesson
MA.K12.MTR. 2.1	Demonstrate understanding by representing problems in multiple ways.	Notes I — two students' plans for one expression are written side by side and both are carried to the end, so the value is seen to be independent of the route.
MA.K12.MTR. 5.1	Use patterns and structure to help understand and connect mathematical concepts.	Notes III and boards 4, 7 and 8 — 8, 4 and 16 are rewritten as powers of 2, which is the only way expressions with different bases can be compared at all.
MA.K12.MTR. 6.1	Assess the reasonableness of solutions.	Board 9 — Xavier's claim is tested rather than believed; the student converts the second expression and finds 3^{-8} .

4. Learning Target and Essential Question

Learning target. I can evaluate an expression with integer exponents using the laws, and decide whether two expressions with different bases are equivalent.

Essential question. How do you determine whether two expressions are equivalent when they do not have the same base?

Building on. 3.01–3.05 — the seven laws with integer exponents and rational bases; Unit 2 — perfect squares and cubes ($8 = 2^3$, $9 = 3^2$, $64 = 2^6 = 4^3 = 8^2$).

Working toward. Thread A (the next two days) — the same laws on variable bases, MA.8.AR.1.1; Unit 4 — operations in scientific notation.

5. Sequence

read from the deck; never typed

Segment	Min	What the teacher does	What students do
Settle in; post the learning target	1	Post the target; say the one line the Teacher's Edition gives for it, and nothing else yet.	Copy the target.
Warm-Up	5	Two minutes silent, then reveal. One sentence of reteach per question at most.	Four retrieval questions, alone, no notes.
Notes	14	Teach from the front. Expand once per law; say the OFF line out loud.	Copy the notes; answer the checks aloud.
Example 1	3	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Example 2	3	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Whiteboards	10	One question at a time. Boards down until the cue, all up together; scan the back row first.	Answer on a board; hold it up on three.
Independent Set	6	Silent. Circulate and mark what you see; do not teach.	Six questions, written, alone.
IXL	5	Assign the day's IXL skills.	Start the skills; finish tonight.
Total	53		

6. Gradual Release

I do. The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the OFF line is said out loud rather than printed.

We do. Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

You do. The nine-question whiteboard round (10 minutes) and then the six-question independent set, silent and written.

7. Higher-Order Questions

Question	DOK
Two expressions have different bases. What has to be true before you are allowed to say they are equivalent?	DOK 2
Xavier's two expressions differ by exactly one in the exponent. Where in his conversion did that one go missing?	DOK 3
Could 3^5 ever be written as a power of 2? Say why or why not.	DOK 3

8. Checks for Understanding

each board, and what to do about it

#	What the board asks	Answer	If the board is wrong
1	$\frac{(-7)^5}{(-7)^0 \cdot (-7)^2}$	-343	343 — sign dropped on an odd power; 'undefined' — $(-7)^0$ in the denominator read as 0 [8.NSO.1.3: zero exponent law, $a^0 = 1$]
2	$\frac{\left(\frac{3}{4}\right)^{-3} \cdot \left(\frac{3}{4}\right)^0}{\left(\frac{3}{4}\right)^{-2}}$	$\frac{4}{3}$	$3/4$ — $(3/4)^{-1}$ left unflipped [negative exponent law]; $(3/4)^{-5}$ or $(4/3)^5$ — subtracted $-3 - 2$ [integer subtraction of a negative]
3	$\frac{(2^{-1})^3 \cdot \left(\frac{1}{2}\right)^{-4}}{2 \cdot 2^2}$	$\frac{1}{4}$	4 — 2^{-2} left unflipped [negative exponent law]; $1/2$ — the denominator $2 \cdot 2^2$ counted as 2^2 [product of powers]
4	One flash drive holds 2^{20} megabytes. Another holds 16^5 megabytes. The first one costs 45 dollars. Which holds more — or is it the same? Show why.	The same: $16^5 = (2^4)^5 = 2^{20}$	' 16^5 holds more' — the bases were compared, not the values [8.NSO.1.3: rewrite to a common base]; ' 2^{20} holds more' — the exponents were compared; 2^9 — added $4 + 5$ in the power-of-a-power step [confused with product of powers]; anything built from 45 — the unneeded figure was used
5	25^7 Which expression is equivalent?	$C — 5^{14}$	(A) squared the exponent, $7^2 = 49$ [the outer exponent applied to the exponent]; (B) replaced 25 by 5 and kept the exponent [the 5^2 was not written]; (D) added $2 + 7$ instead of multiplying [power of a power confused with product of powers]
6	$\left(\frac{1}{4}\right)^4 \cdot 2^{-5}$ Which expression is equivalent?	$A — 2^{-13}$	(B) dropped both minus signs [negative exponent law: a negative exponent does not disappear]; (C) wrote $1/4$ as 2^{-1} instead of 2^{-2} [the base 4 is 2^2 , not 2]; (D) multiplied $-8 \cdot 5$ [confused with power of a power]
7	$2^5 \cdot 4^8$	2^{21}	8^{13} — multiplied the bases and added the exponents [7.NSO.1.1 misconception: operating on the bases]; 2^{16} — 4^8 rewritten but the 2^5 dropped
8	$4^7 \cdot 4^{-2}$	2^{10}	2^5 — 4 replaced by 2 without the exponent 2; 4^9 or 2^{18} — subtracted instead of adding a negative
9	Xavier says $(1/9)^2 \cdot 3^{-5}$ and $(1/3)^6 \cdot 9^{-1}$ are equivalent because both can be written as 3^{-9} . Do you agree?	No — the second is 3^{-8}, not 3^{-9}	'Yes' — 9^{-1} rewritten as 3^{-1} [the base 9 is 3^2]

MTR.4.1 — the one peer move. When a board splits — no option holding about two-thirds of the room — sixty seconds of “convince the person next to you”, then re-vote, then reveal. The minute comes out of the whiteboard remainder.

9. Differentiation

ESE / IEP. Give the powers of 2, 3 and 5 from Unit 2 as a printed strip — 2^1 to 2^{10} , 3^1 to 3^5 , 5^1 to 5^4 — and let it stay out. Rewriting a base is a lookup today, not a recall.

ELL. Equivalent is the word being tested. Write EQUIVALENT = same value, different form with $8^5 = 2^{15}$ under it, and ask 'same value?' on every board that shows two expressions.

Enrichment. Ask whether 4^6 and 8^4 are equivalent and how the student knows, then for two expressions with different bases that both equal 2^6 .

10. Closure, Vocabulary, Materials, Practice

Closure. Board 9 is written work and it is the exit evidence: the student has to convert the second expression and name the exponent it really has. Read the boards, not the papers.

Vocabulary. **common base; equivalent**

Materials. Whiteboards and markers. Calculators are allowed on everything and are how students confirm $8^5 = 2^{15} = 32,768$ once.

Practice. The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, due at the start of the next class: Evaluate expressions using properties of exponents — 6T7; Identify equivalent expressions involving exponents II — QDM.